



MATERIALS SCIENCE & PRECAST DURABILITY

Ultra-High-Performance Concrete (UHPC) in High-Salinity Transit Ports

Durability modeling, chloride ingress resistance, and accelerated steam curing regimes for heavy container terminals.

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EXECUTIVE TECHNICAL SUMMARY

In high-throughput intermodal dry ports and maritime container terminals, concrete infrastructure is exposed to relentless cyclic axle loads (up to 42 MT from container reach stackers) coupled with aggressive saline dust and chemical runoffs. This study evaluates Asterra's precast UHPC pavement slabs, demonstrating over 160 MPa compressive strength and a two-order-of-magnitude reduction in chloride diffusion relative to standard maritime concrete mixes.

Key Engineering Findings & Takeaways

- **Compressive & Flexural Superiority:** Dense reactive-powder matrix combined with brass-coated high-carbon micro-steel fibers delivers 165 MPa compressive strength and 22.4 MPa flexural tensile resistance.
- **Impermeable Matrix:** Water absorption is below 0.2%, reducing chloride penetration coefficients to less than $1.0 \times 10^{-13} \text{ m}^2/\text{s}$ and virtually eliminating rebar corrosion risk.
- **Accelerated Steam-Curing:** Computerized thermal steam curing at 85°C for 48 hours achieves 90% of ultimate 28-day strength in 72 hours, drastically shortening terminal paving turnaround times.

1. The Durability Crisis in Maritime Terminals

Traditional container port pavements constructed with conventional C35/45 or C40/50 ready-mix concrete suffer premature surface spalling and micro-cracking under heavy industrial wheel loading. In dry port and maritime environments, surface cracks allow rapid penetration of chloride ions and sulfates, initiating rapid electrochemical corrosion of reinforcing steel and requiring costly operational shutdowns every 5 to 7 years.

2. Precast UHPC Formulation & Steam Curing

Asterra's Division 02 engineered an optimized particle packing curve incorporating micro-silica fume, ground quartz flour, and high-range polycarboxylate superplasticizers at a water-binder ratio below 0.18. The concrete is reinforced with 2.5% by volume brass-coated micro-steel fibers, providing strain-hardening ductile post-crack behavior.

PERFORMANCE CHARACTERISTIC	STANDARD C35/45 CONCRETE	HIGH-STRENGTH C60 CONCRETE	ASTERRA PRECAST UHPC-160
28-Day Compressive Strength	45 MPa	62 MPa	165 MPa
Flexural Tensile Strength	5.2 MPa	7.8 MPa	22.4 MPa
Water Absorption Capacity	4.8%	2.9%	< 0.2%
Chloride Diffusion (m²/s)	1.2 × 10 ⁻¹¹	4.5 × 10 ⁻¹²	< 1.0 × 10 ⁻¹³
Expected Service Life (Saline)	20 – 25 Years	35 – 40 Years	100+ Years

Verified Reference Project: Modjo Dry Port Heavy Logistics Pavements

Client: Ethiopian Shipping and Logistics Services Enterprise (ESLSE) // **Scope:** 85,000 m² of precision precast interlocking heavy-duty pavement slabs designed for 42-ton axle reach stackers. Zero slab cracking or joint deterioration recorded under continuous commercial operations.